

EDUNET AICTE INTERNSHIP

AI AGENT FOR CHRONIC DISEASE MONITORING

Presented By:

Student name : ABISHEK.S

**College Name & Department : Global Institute of Engineering And Technology-Vellore
B.Tech Information Technology**

OUTLINE

- Problem Statement
- Technology used
- Wow factor
- End users
- Result
- Conclusion
- Git-hub Link
- Future scope
- IBM Certifications

PROBLEM STATEMENT

An AI agent for chronic disease monitoring helps patients and healthcare providers manage long-term conditions effectively. It continuously analyzes health data from wearables, medical records, and patient inputs to detect early warning signs. Using AI and predictive analytics, it offers personalized insights, medication reminders, and lifestyle recommendations. The agent supports diseases like diabetes, hypertension, and heart conditions with real-time monitoring and alerts. It enables proactive care, reduces hospital visits, and improves patient adherence to treatment plans. This intelligent assistant bridges the gap between patients and providers, enhancing chronic care outcomes.

Proposed Solution:

Develop an AI-powered agent that continuously monitors chronic diseases by analyzing data from wearables, medical records, and patient inputs. It provides real-time alerts, personalized health insights, medication reminders, and lifestyle recommendations for conditions like diabetes, hypertension, and heart diseases.

TECHNOLOGY USED

IBM cloud lite services

Natural Language Processing (NLP)

Vector Indexing

Artificial Intelligence

IBM Granite model

IBM CLOUD SERVICES USED

- IBM Cloud Watsonx AI Studio
- IBM Cloud Watsonx AI runtime
- IBM Cloud Agent Lab
- IBM Granite foundation model
- Cloud Object Storage

WOW FACTORS

- This agent will revolutionize chronic disease care by enabling real-time health tracking, reducing hospital admissions, improving treatment adherence, and empowering both patients and providers through intelligent, personalized insights.

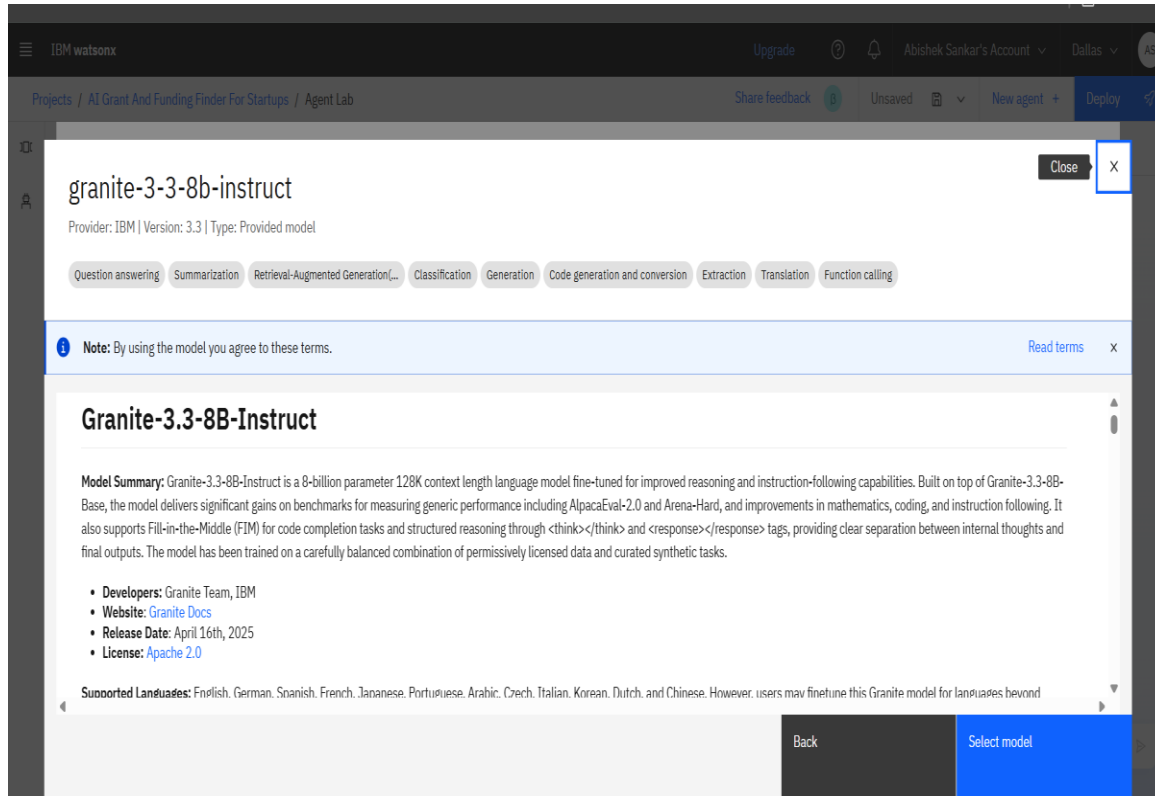
Unique Features:

- Continuous monitoring via wearable and health record integration
- Personalized recommendations for medication, diet, and lifestyle
- Real-time alerts for critical health deviations
- Predictive analysis for early detection of complications
- Daily symptom tracking with natural language input
- Smart reminders for medication and health check-ins
- Secure data sharing with healthcare providers for better collaboration

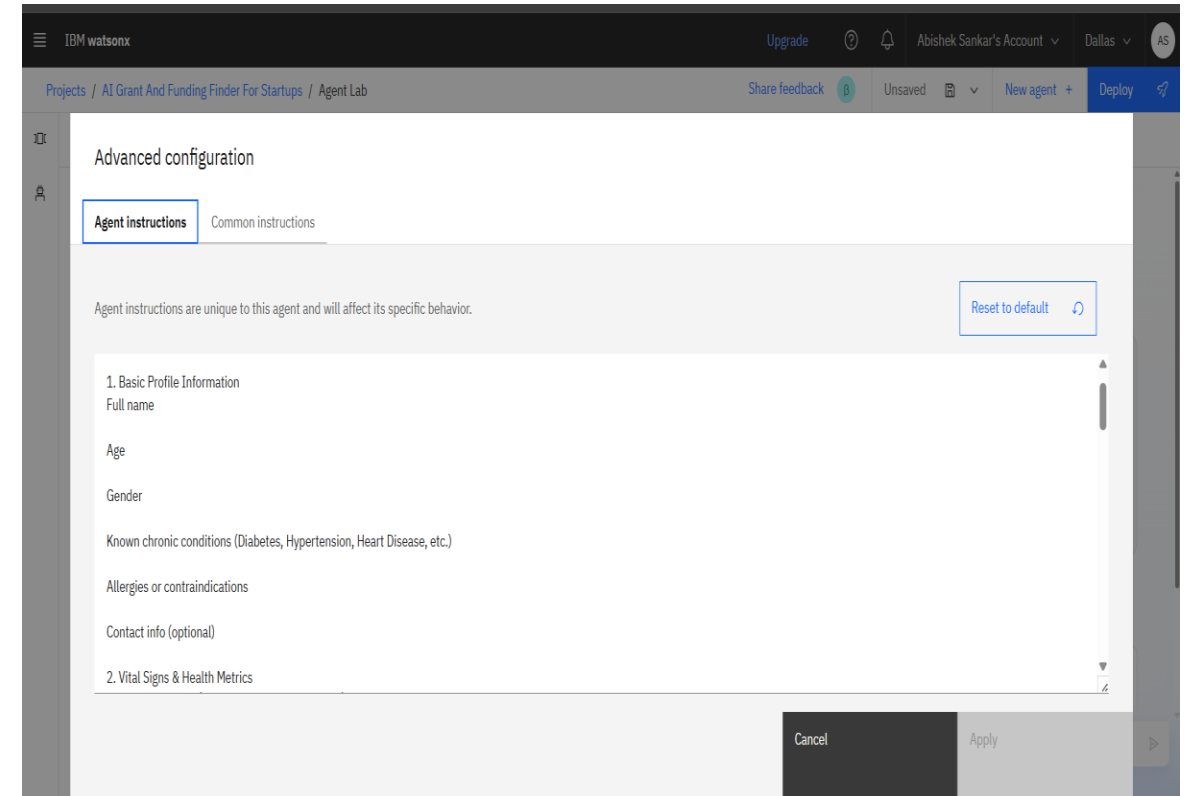
END USERS

- Patients with chronic conditions (e.g., diabetes, hypertension, heart disease)
- Caregivers and family members
- Healthcare Providers (Doctors, Nurses, Specialists)
- Hospitals and Clinics
- Telemedicine platforms
- Health insurance companies
- Government and public health agencies

RESULTS

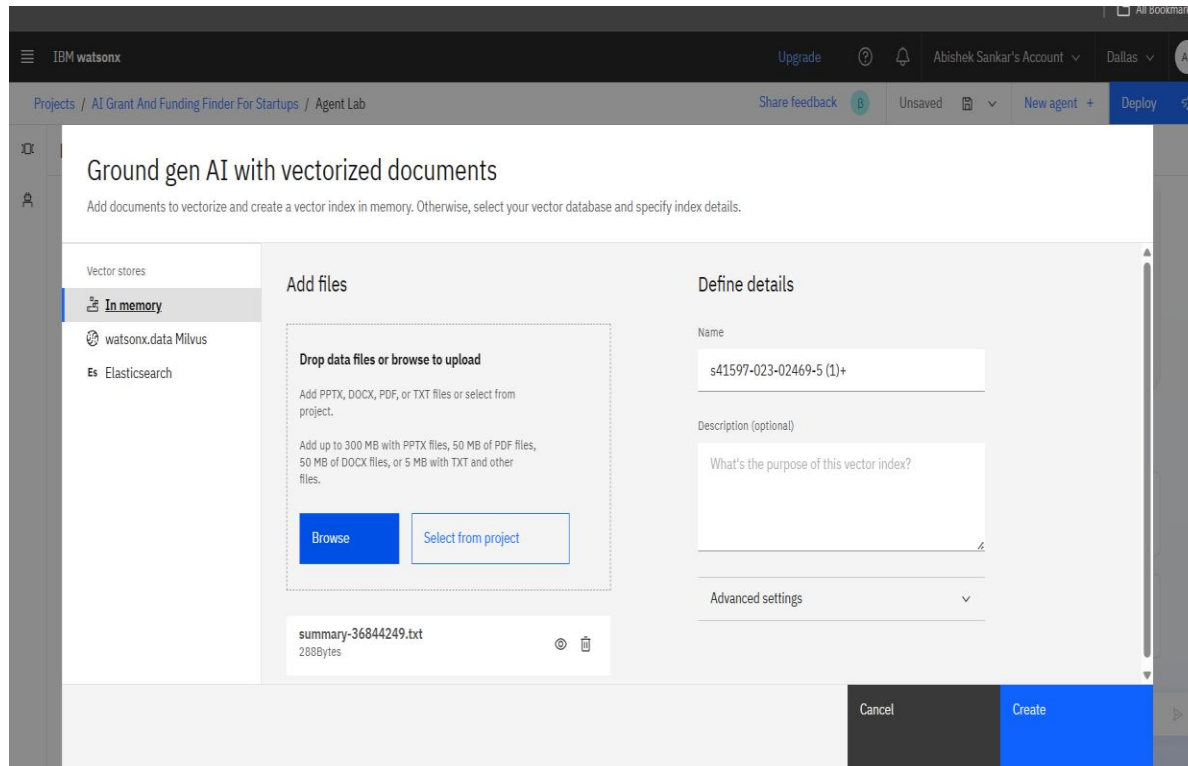


Using Granite Model

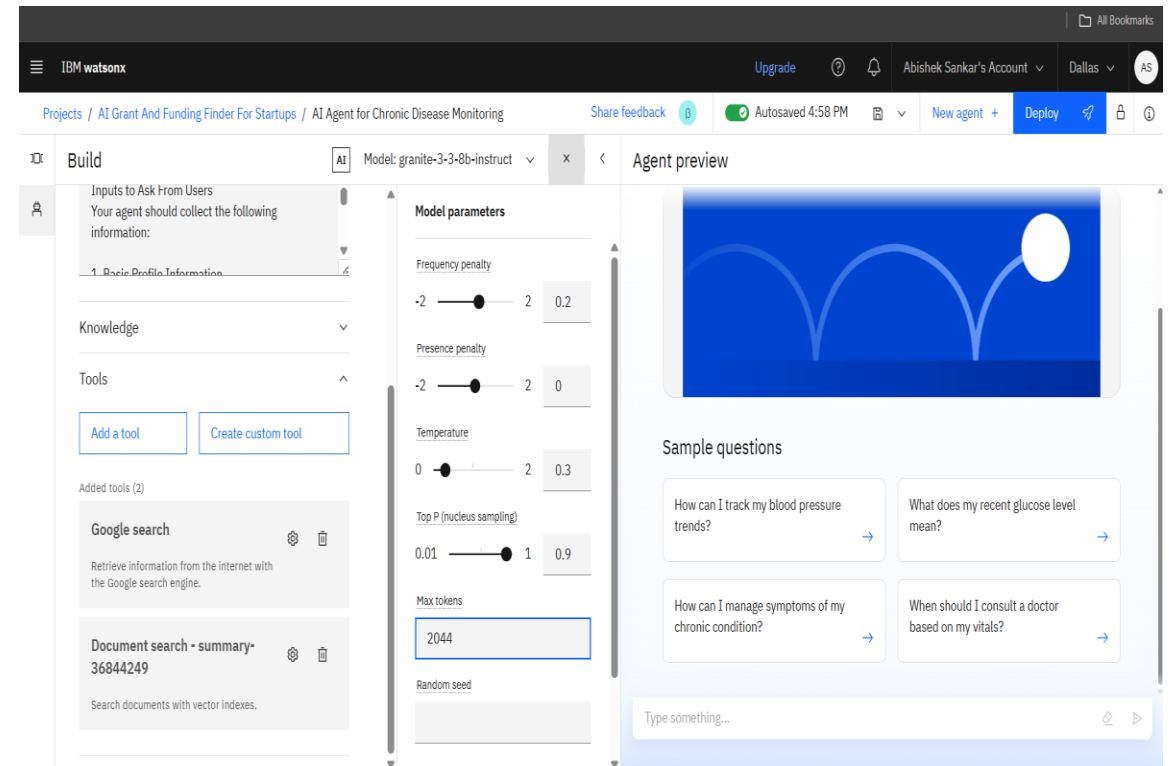


Adding Instructions

RESULTS



Vectorized Documents



Model Parameters

RESULTS

The screenshot shows the IBM Watsonx interface for the 'AI Agent for Chronic Disease Monitoring'. The 'Deployments' tab is active, displaying a table with one deployment. The deployment is named 'AI Agent for Chronic Disease Monitoring', is of type 'Online', and has a status of 'Deployed' (indicated by a green checkmark). It was last modified '32 seconds ago' by 'Abishek Sankar (You)'. The asset type is 'Ai service' and it has a tag 'wx-agent'.

Name	Type	Status	Asset	Asset type	Tags	Last modified
(y) AI Agent for Chronic Disease Monitoring	Online	Deployed	AI Agent for Chronic Disease Monitoring	Ai service	wx-agent	32 seconds ago Abishek Sankar (You)

Agent Deployed

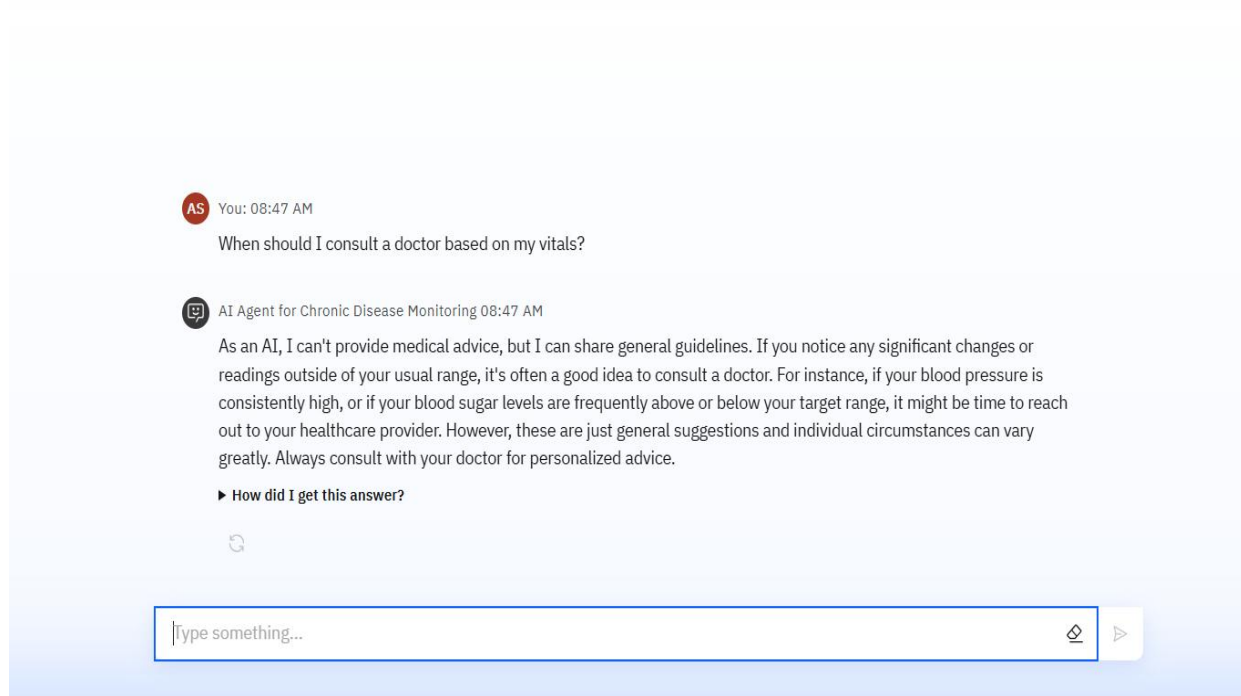
The screenshot shows the 'Preview' tab of the 'AI Agent for Chronic Disease Monitoring' interface. It displays 'Quick start samples' with four sample questions, each with a right arrow button:

- How can I track my blood pressure trends?
- What does my recent glucose level mean?
- How can I manage symptoms of my chronic condition?
- When should I consult a doctor based on my vitals?

At the bottom, there is a text input field with the placeholder 'Type something...' and a right arrow button.

Sample Questions

RESULTS



AS You: 08:47 AM

When should I consult a doctor based on my vitals?

AI Agent for Chronic Disease Monitoring 08:47 AM

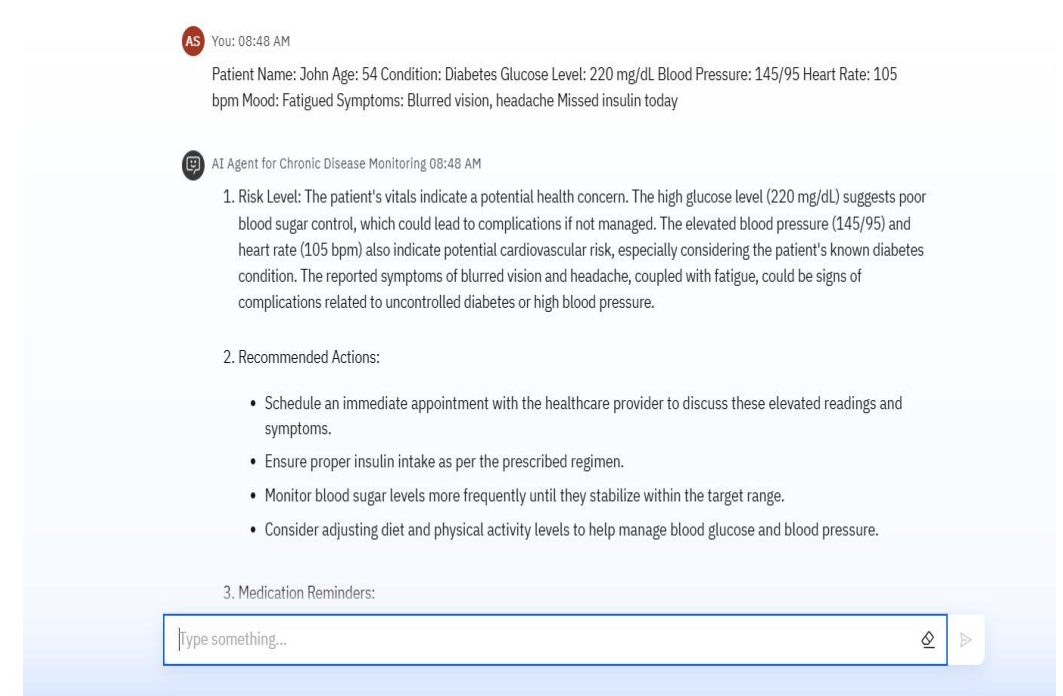
As an AI, I can't provide medical advice, but I can share general guidelines. If you notice any significant changes or readings outside of your usual range, it's often a good idea to consult a doctor. For instance, if your blood pressure is consistently high, or if your blood sugar levels are frequently above or below your target range, it might be time to reach out to your healthcare provider. However, these are just general suggestions and individual circumstances can vary greatly. Always consult with your doctor for personalized advice.

► How did I get this answer?

🔄

Type something...

Answer Generated For Sample Question



AS You: 08:48 AM

Patient Name: John Age: 54 Condition: Diabetes Glucose Level: 220 mg/dL Blood Pressure: 145/95 Heart Rate: 105 bpm Mood: Fatigued Symptoms: Blurred vision, headache Missed insulin today

AI Agent for Chronic Disease Monitoring 08:48 AM

1. Risk Level: The patient's vitals indicate a potential health concern. The high glucose level (220 mg/dL) suggests poor blood sugar control, which could lead to complications if not managed. The elevated blood pressure (145/95) and heart rate (105 bpm) also indicate potential cardiovascular risk, especially considering the patient's known diabetes condition. The reported symptoms of blurred vision and headache, coupled with fatigue, could be signs of complications related to uncontrolled diabetes or high blood pressure.

2. Recommended Actions:

- Schedule an immediate appointment with the healthcare provider to discuss these elevated readings and symptoms.
- Ensure proper insulin intake as per the prescribed regimen.
- Monitor blood sugar levels more frequently until they stabilize within the target range.
- Consider adjusting diet and physical activity levels to help manage blood glucose and blood pressure.

3. Medication Reminders:

Type something...

Customized User Input and Generated Result

CONCLUSION

- ❖ The agent delivers personalized insights, monitors patient vitals in real-time, and provides timely alerts to prevent complications.
- ❖ It enhances chronic care by automating follow-ups, managing health data, and supporting informed clinical decisions.
- ❖ Health AI agents boost efficiency, adherence, and proactive care across both individual and institutional healthcare systems.

GITHUB LINK

LINK : <https://github.com/ABISHEK-S-15/edunet-intern.git>

FUTURE SCOPE

- Multilingual Health Support for regional and global accessibility
- Voice – Enabled Health Interaction for elderly and visually impaired patients
- Real-Time Caregiver Collaboration and shared dashboards
- Predictive Risk Modeling for emerging comorbidities
- Integration with Telemedicine and HER Systems
- AI-Assisted Report Generation for clinicians and insurance claims

IBM CERTIFICATIONS



IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

ABISHEK S

for the completion of

**Lab: Retrieval Augmented Generation with
LangChain**

(ALM-COURSE_3824998)

According to the Adobe Learning Manager system of record

Completion date: 16 Jul 2025 (GMT)

Learning hours: 20 mins

In recognition of the commitment to achieve
professional excellence



Abishek S

Has successfully satisfied the requirements for:

Journey to Cloud: Envisioning Your Solution



Issued on: Jul 18, 2025

Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/a77204ff-895f-49c7-bcef-3b260b95d62d>





THANK YOU